

Barrovian Metamorphic Zones, Isograd Concepts & Pelitic Facies

Authoritative Reference: John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapters 21 & 28)

Comprehensive Chapter Masterclass: Metamorphism of Pelitic Sediments

Barrow's Classical Zones, The Physical Concept of Isograds, AKF/AFM Topologies & Devolatilization Reactions

1. Historical Foundations: George Barrow & The Scottish Highlands

The systematic zoning of regional metamorphic terranes originated with the pioneering field work of **George Barrow (1893, 1912)** in the Dalradian Supergroup of the Grampian Highlands, Scotland. Barrow mapped a sequence of systematic mineralogical transformations occurring within metamorphosed pelitic rocks (shales, mudstones, and siltstones) progressing outward from unmetamorphosed or low-grade sediments toward deeper, hotter orogenic infrastructure.

"In a pioneering study of regional metamorphism, George Barrow (1893, 1912) made one of the first and most lasting contributions to our understanding of the progressive nature of metamorphism... Barrow noted significant and systematic mineralogical changes in the pelitic rocks as he progressed from the coastal slates to the more deeply eroded interior of the metamorphic belt. He recognized six progressive mineral zones, each based on the first appearance of a distinctive new index mineral."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 21, p. 486)

The classical sequence of index minerals recognized by Barrow defines what is now universally termed the **Barrovian metamorphic series** (representing typical medium *P/T* regional orogenic metamorphism):

1. **Chlorite Zone:** Very low to low grade; rocks are slates and phyllites. Typical assemblage: *Chlorite + Muscovite + Quartz + Albite*.
2. **Biotite Zone:** Low grade (greenschist facies); phyllites and fine schists. Typical assemblage: *Biotite + Chlorite + Muscovite (phengitic) + Quartz + Albite*.
3. **Garnet Zone:** Low-medium grade (upper greenschist to epidote-amphibolite); schists characterized by almandine-rich garnet. Assemblage: *Garnet + Biotite + Muscovite + Chlorite + Quartz + Plagioclase*.
4. **Staurolite Zone:** Medium grade (lower amphibolite facies); schists marked by the appearance of staurolite.
5. **Kyanite Zone:** Upper-medium grade (middle amphibolite facies); schists bearing kyanite, indicating substantial lithostatic overburden (medium to high pressure, typical of crustal thickening).

6. **Sillimanite Zone:** High grade (upper amphibolite to granulite facies); rocks contain fibrous sillimanite (fibrolite) or prismatic sillimanite, coexisting with biotite, muscovite, quartz, and plagioclase (and at the highest grades, partial melting begins via muscovite breakdown).

area, Barrow noted significant and systematic mineralogical changes in the *pelitic* rocks (originally shales). He found that he could subdivide the area into a series of **metamorphic zones** (Figure 8), each based on the appearance of a new mineral in the metamorphosed pelitic rocks as metamorphic

to as the **Barrovian zones**. The $P-T$ conditions represented (see the range of estimated metamorphic field gradients in Figure 1) are also referred to as **Barrovian-type** (or style) metamorphism, which is fairly typical of many orogenic metamorphic belts. C. E. Tilley (1925) and W. Q. Kennedy

FIGURE 8 Regional metamorphic map of the Scottish Highlands, showing the zones of minerals that develop with increasing metamorphic grade. From Gillen (1982).

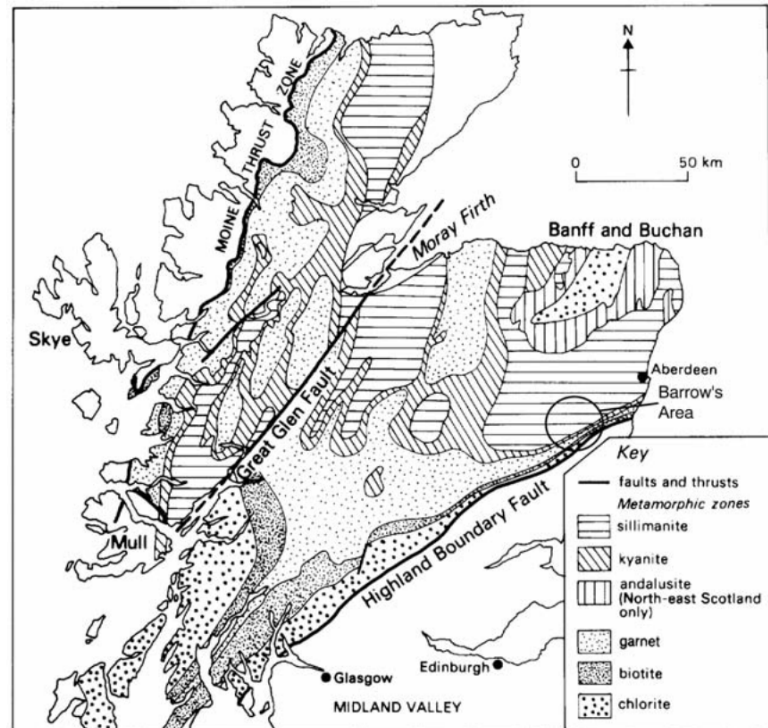


Figure 21-8 from John D. Winter (p. 486): Regional metamorphic map of the Scottish Highlands showing the classic Barrovian metamorphic zones mapped by George Barrow, along with the lower-pressure Buchan facies series (Buchan/Banff district) characterized by cordierite and andalusite.

2. The Physical Meaning of an Isograd & Modern Chemographic Analysis

In 1925, **Cecil Edgar Tilley** introduced the term **isograd** (from the Greek *iso* = equal, *gradus* = grade or step) to describe the mappable boundary that separates two metamorphic zones. In classical mapping:

- An **isograd** represents the surface of first appearance of an index mineral in a rock of appropriate bulk composition (most notably, pelites).
- Because temperature and pressure vary continuously across three-dimensional space, the mapped isograd line on a geological map represents the **intersection of an isogradic surface with the Earth's topographic surface**.
- When crossing an isograd in the field (e.g., crossing the biotite isograd), one enters the corresponding zone (the biotite zone). The zone shares the name of its defining isograd.

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 21, p. 487)

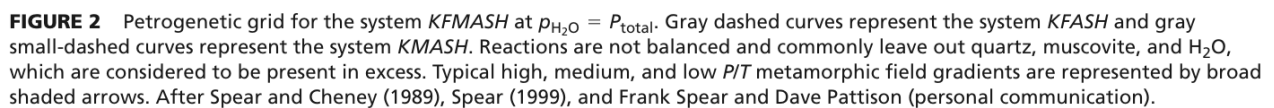


Figure 28-2 from John D. Winter (p. 645): Petrogenetic grid for pelites in the system KFMASH (K_2O – FeO – MgO – Al_2O_3 – SiO_2 – H_2O) across metamorphic grades, displaying dehydration equilibria, the Al_2SiO_5 triple point, and continuous solid-solution reactions.

3. Detailed Petrology of the Biotite Isograd & AKF/AFM Topology

In low-grade pelitic rocks (slates and phyllites of the chlorite zone), the primary sheet silicates are **chlorite** and **phengitic muscovite** (a white mica rich in the celadonite component, where Al in octahedral and tetrahedral sites is substituted by (Mg, Fe) + Si: $\text{Al}^{\text{VI}} + \text{Al}^{\text{IV}} \leftrightarrow (\text{Mg, Fe})^{\text{VI}} + \text{Si}^{\text{IV}}$).

As temperature increases into the range of approximately **300°C to 400°C** at crustal pressures of 2 to 6 kbar, the first brown biotite flakes nucleate in the rock. The specific reaction responsible for the appearance of biotite depends on the bulk potassium content and whether K-feldspar or phengite acts as the potassium reservoir:

Reaction 1: The Discontinuous Biotite-In Reaction (K-Feldspar Bearing Pelites)

In pelites containing detrital or diagenetic K-feldspar, biotite forms at the lowest temperatures by the univariant reaction in the KFMASH/KFMASH system:



This reaction consumes K-feldspar completely in most normal aluminous pelites, explaining why K-feldspar is typically absent from intermediate-grade Barrovian pelites until the high-grade second-sillimanite isograd is reached.

Reaction 2: The Continuous Phengite Breakdown Reaction (K-Feldspar Free Pelites)

In common K-feldspar-free shales, the potassium is supplied by the phengite component of white mica. As temperature rises, the phengitic muscovite sheds its Fe-Mg component to react with chlorite:



Because this is a continuous multi-component reaction involving Fe-Mg-Al solid solutions, chlorite does not disappear abruptly at the isograd; instead, chlorite and biotite coexist across the entire width of the biotite zone!

phases. In reduced system diagrams, such as the *AKF* and *AFM* diagrams, additional phases from the full *KFMASH* system may also be required to balance a reaction. In this case Qtz and H₂O (in parentheses) are present and needed to balance Reaction (2), but do not plot on the projected compatibility diagrams. Occasionally I shall ignore such non-diagram phases in reactions in order to concentrate on the relationship between reactions and tie-line geometry in the diagrams. If a reaction conforms to these geometric “rules” we shall generally content ourselves with the knowledge that it can be balanced, without formally doing so.

Reaction (2) can be demonstrated geometrically by resorting to the *AKF* projection (Figure 3). At grades below this biotite–isograd reaction, Chl and Kfs are stable together, as indicated in Figure 3a by the tie-lines connecting them. Above the isograd (Figure 3b), the new Bt–phengitic Ms tie-line separates Chl and Kfs, so that Bt + Ms are stable together and Chl + Kfs are not, as Reaction (2) indicates. Remember that crossing tie-lines in compatibility diagrams may reflect a reaction relating the minerals that plot at the ends of the crossed lines. The replacement of the Chl–Kfs tie-line with the crossing Bt–Ms tie-line (a “tie-line flip”) thus reflects Reaction (2). Note from the full reaction above: if muscovite or quartz is absent, or H₂O is scarce, biotite would be stable at lower temperatures (biotite plots on Figure 3a, but natural pelite compositions are generally too Al rich for it to develop). Quartz and muscovite are abun-

A reaction will occur only in rocks that plot within the quadrilateral having the reactants and products as corners. This is a major weakness of *P-T* grids. Good ones show all of the important univariant reactions for all compositions in the system. Any single rock, however, will manifest but a few of these reactions. That’s why we compare grids to compatibility diagrams and, as we will see in a moment, pseudosections.

Although Reaction (2) appears to be a *discontinuous* reaction in Figure 3b, the temperature of the isograd varies with the Fe/Mg ratio (Fe-rich biotite becomes stable at lower temperatures). At metamorphic grades above the tie-line flip reaction in Figure 3, the composition of the white mica that coexists with biotite and chlorite gradually becomes less phengitic via *continuous* reactions by which the most Al-poor white mica breaks down, and the amounts of chlorite and biotite increase (as they become more Al- and Mg-rich). This causes the Ms–phengite solid-solution range (line) to get shorter in the *AKF* diagram as metamorphic grade increases, and the Ms–Chl–Bt and Ms–Bt–Kfs sub-triangles respond by migrating toward the A apex in Figure 4. The migration of the Chl–Bt–Ms triangle gradually includes more of the shaded pelite composition range, so that biotite begins to appear in progressively more aluminous pelite compositions as a result of the *continuous* reactions, not a *discontinuous* reaction, such as (2), which we tend to associate in our minds with isograds. This is another way of illustrating the point

Figure 28-3 from John D. Winter (p. 646): *AKF* compatibility diagrams ($A = \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3 - (\text{Na}_2\text{O} + \text{K}_2\text{O} + \text{CaO})$, $K = \text{K}_2\text{O}$, $F = \text{FeO} + \text{MgO} + \text{MnO}$) projected from quartz and H₂O. (a) Sub-biotite zone: Chlorite and K-feldspar are stable together. (b) Above the biotite isograd: The new Bt–Ms tie-line forms, breaking the Chl–Kfs tie-line and creating the stable assemblage **Biotite + Chlorite + Muscovite + Quartz**.

Therefore, in the biotite zone of a Barrovian sequence, the classic equilibrium mineral assemblage is:

Biotite + Chlorite + Phengitic Muscovite + Quartz + Albite (± epidote)



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 4

Question Statement

Q.4: In a Barrovian zone of metamorphism, presence of biotite + chlorite + phengitic muscovite + quartz + albite assemblage marks the:

(a) Albite zone

(b) Biotite zone

(c) Zeolite zone

(d) Chlorite zone

Correct Answer: (b) Biotite zone



The first appearance of metamorphic biotite in equilibrium with chlorite, phengitic muscovite, quartz, and albite definitively marks the entry into the Barrovian Biotite Zone.

Exhaustive Option-by-Option Critical Analysis

(b) Biotite zone

CORRECT

Why it is uniquely correct: In the classic Barrovian progressive metamorphic sequence established by Barrow (1893, 1912) and codified in petrology textbooks (Winter, 2010, p. 486; Yardley, 1989; Bucher & Grapes, 2011), the incoming of brown biotite in pelites defines the **biotite isograd** and the beginning of the **biotite zone**. Because the reaction forming biotite in common pelites involves the reaction of chlorite with the phengitic component of muscovite (or K-feldspar), chlorite does not disappear immediately but persists alongside newly formed biotite. Therefore, the diagnostic 5-phase equilibrium assemblage of the biotite zone is specifically *biotite + chlorite + phengitic muscovite + quartz + albite*.

(a) Albite zone

INCORRECT

Why it is incorrect: There is no such division as an "Albite zone" in the Barrovian metamorphic series. While albite is the stable sodic plagioclase feldspar throughout low-grade metamorphism (persisting across both the chlorite and biotite zones up to the peristerite solvus around 450°C in the oligoclase/garnet zone), it is an ubiquitous matrix phase rather than a defining index mineral. Using albite as an index mineral name violates standard IUGS metamorphic nomenclature.

(c) Zeolite zone

INCORRECT

Why it is incorrect: The *Zeolite facies* (or zeolite zone) represents very low-grade burial metamorphism and diagenesis (temperatures below ~200°C–250°C and pressures < 3 kbar), characterized by hydrated calcium-sodium aluminosilicates such as heulandite, analcime, and laumontite in volcanoclastic rocks, or smectite/illite in pelites. Biotite is completely unstable under zeolite facies conditions; true biotite requires thermal conditions above ~300°C to nucleate in pelitic compositions.

(d) Chlorite zone

INCORRECT

Why it is incorrect: In the *Chlorite zone* (the lowest-grade zone of the Barrovian sequence), the rock is composed exclusively of *chlorite + muscovite (illite/phengite) + quartz + albite*. By strict definition, the chlorite zone lies **below the biotite isograd**; biotite is completely absent. The moment the very first grains of biotite appear in the thin section or hand specimen, the rock has transitioned across the isograd into the biotite zone!



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Complete Facies Equivalences, Field Recognition & 8 Exam-Grade Multiple-Choice Questions

Barrovian Metamorphic Sequence: Mineral Zones & Facies Equivalents

Barrovian Zone	Defining Isograd Reaction (in Pelites)	Diagnostic Equilibrium Assemblage	Approx. T (°C)	Metamorphic Facies
Chlorite Zone	Smectite/Illite → Chl + Ms	Chlorite + Muscovite + Albite + Quartz ± Epidote	< 300°C	Sub-greenschist to Lower Greenschist
Biotite Zone	Chl + Phengite / Kfs → Bt + Ms + Qtz + H ₂ O	Biotite + Chlorite + Muscovite + Quartz + Albite	300 – 420°C	Middle Greenschist Facies
Garnet Zone	Chl + Ms + Qtz → Grt (Alm) + Bt + H ₂ O	Almandine + Biotite + Muscovite + Quartz + Oligoclase	420 – 520°C	Upper Greenschist to Epidote-Amphibolite
Staurolite Zone	Grt + Chl + Ms → St + Bt + Qtz + H ₂ O	Staurolite + Biotite + Almandine + Muscovite + Quartz + Plagioclase	520 – 580°C	Lower Amphibolite Facies
Kyanite Zone	St + Chl + Ms → Ky + Bt + Qtz + H ₂ O	Kyanite + Staurolite + Biotite + Almandine + Muscovite + Quartz + Plagioclase	580 – 650°C	Middle Amphibolite Facies
Sillimanite Zone	Ky → Sil (polymorphic) & Ms + Qtz → Sil + Kfs + Melt	Sillimanite (fibrolite/prismatic) + Biotite + Garnet + Plagioclase + Quartz ± K-feldspar	> 650°C	Upper Amphibolite to Granulite Facies

Diagnostic Petrographic Criteria: Identifying the Biotite Zone

- **Microscopic Texture:** Rocks of the biotite zone are typically fine-grained phyllites or very fine schists. Brown pleochroic biotite flakes (often sub-millimeter in size) lie along the continuous cleavage planes (\$) defined by aligned muscovite and chlorite.
- **Coexistence of Chlorite and Biotite:** Green chlorite flakes remain abundant and intimately intergrown with brown biotite, demonstrating chemical equilibrium across the divariant AFM field.

- **Phengite Substitution:** Muscovite shows elevated Si:Al ratios (phengitic character), reflecting moderate-to-high pressure conditions during orogenic thickening.
- **Plagioclase Composition:** Plagioclase remains strictly pure albite (\$ to \$). Calcification across the peristerite solvus into oligoclase occurs only at higher grades near the garnet isograd.

8 High-Yield Practice Questions on Pelitic Metamorphism & Barrovian Zones

1. In the Dalradian rocks of the Scottish Highlands, who first introduced the concept of an "isograd" to delineate metamorphic zones?

- (a) George Barrow (b) C. E. Tilley (c) Pentti Eskola (d) Arthur Holmes

Answer: (b) C. E. Tilley — While George Barrow (1893) first mapped the mineral zones, Cecil Edgar Tilley (1925) coined and defined the term "isograd" as a line of equal metamorphic grade based on the first appearance of an index mineral.

2. Which of the following is the correct ascending order of Barrovian metamorphic zones in pelitic rocks?

- (a) Chlorite → Biotite → Staurolite → Garnet → Kyanite → Sillimanite
 (b) Chlorite → Biotite → Garnet → Staurolite → Kyanite → Sillimanite
 (c) Chlorite → Garnet → Biotite → Staurolite → Sillimanite → Kyanite
 (d) Biotite → Chlorite → Garnet → Kyanite → Staurolite → Sillimanite

Answer: (b) — The standard Barrovian sequence is Chlorite → Biotite → Garnet → Staurolite → Kyanite → Sillimanite.

3. Why is K-feldspar typically absent in intermediate-grade Barrovian pelitic schists (biotite to kyanite zones)?

- (a) Potassium is insoluble in metamorphic fluids
 (b) K-feldspar is consumed at the biotite isograd by reaction with chlorite to form biotite + muscovite
 (c) K-feldspar only forms in basaltic rocks
 (d) K-feldspar transforms into albite at low temperatures

Answer: (b) — The reaction $3 \text{Chl} + 8 \text{Kfs} = 5 \text{Bt} + 3 \text{Ms} + 9 \text{Qtz} + 4 \text{H}_2\text{O}$ completely consumes all modal K-feldspar in normal pelites at the onset of the biotite zone.

4. The Buchan (or Abukuma) metamorphic series differs from the Barrovian series because it develops under:

- (a) Higher pressure at similar temperature (b) Lower pressure and higher geothermal gradient (c) Ultra-high pressure subduction (d) Anhydrous conditions

Answer: (b) Lower pressure and higher geothermal gradient — Buchan facies series is characterized by low P/T conditions where andalusite and cordierite develop instead of kyanite and almandine.

5. In pelitic rocks, the change in plagioclase composition from albite ($<An_5$) to oligoclase ($An_{15}-An_{25}$) across the peristerite solvus typically coincides with the:

- (a) Chlorite isograd (b) Biotite isograd (c) Garnet isograd (d) Second sillimanite isograd

Answer: (c) Garnet isograd — The jump across the peristerite solvus from albite to oligoclase occurs at the transition from upper greenschist to epidote-amphibolite/amphibolite facies, near the garnet isograd.

6. In the AFM projection (Thompson, 1957) for pelitic rocks, the phases from which the composition is projected are:

- (a) Albite, Quartz, and CO_2 (b) Muscovite (or K-feldspar), Quartz, and H_2O (c) Calcite, Dolomite, and Quartz (d) Garnet, Biotite, and Staurolite

Answer: (b) Muscovite (or K-feldspar), Quartz, and H_2O — The AFM projection simplifies the 6-component KFMASH system into a 2D ternary diagram by projecting from Muscovite, Quartz, and H_2O (or K-feldspar at higher grades).

7. Phengitic muscovite in low-grade pelitic rocks is characterized by:

- (a) Depletion in silicon and enrichment in sodium
(b) Substitution of $(Mg, Fe) + Si$ for Al in both octahedral and tetrahedral sites: $AlVI + AlIV \leftrightarrow (Mg, Fe)VI + SiIV$
(c) Complete replacement of K by Ca
(d) High fluorine content

Answer: (b) — Phengite is a solid solution between muscovite and celadonite, enriched in Si (up to 3.5 per formula unit) and (Mg, Fe) at higher pressures and lower temperatures.

8. The "second sillimanite isograd" in high-grade pelitic rocks represents:

- (a) The polymorphic transformation of andalusite to sillimanite
(b) The dehydration breakdown of muscovite: $Muscovite + Quartz = Sillimanite + K-feldspar + H_2O$ (or melt)
(c) The breakdown of garnet to staurolite
(d) The first appearance of fibrolite from kyanite

Answer: (b) — The second sillimanite isograd marks the upper stability limit of muscovite in the presence of quartz, producing K-feldspar + sillimanite and initiating anatectic melting.